Question 9: In case of well-defined utility function, show that

$$\textbf{Concavity} \underset{\not\leftarrow}{\rightarrow} \textbf{Quasi-concavity}$$

Answer:

In case of well-defined utility function, concavity implies quasi-concavity but the converse relation is not necessarily true.

$$\text{Concavity} \underset{\not\leftarrow}{\rightarrow} \text{Quasi-concavity}$$

If a function $f(\cdot)$ is concave, then for any two points x, y ∈ X,

$$f(\alpha x + (1-\alpha)y) \geq \alpha f(x) + (1-\alpha)f(y)$$
$$\geq \min\{f(x), f(y)\}$$

for all $\alpha \in (0,1)$.

where the first inequality follows from the definition of concavity, and the second holds true for all concave functions, as depicted in next figure. From the first and third element in the above inequality, we obtain that $f(\alpha x + (1-\alpha)y) \geq \min\{f(x), f(y)\}$, which coincides with the definition of quasi-concavity. As a consequence, concavity implies quasi-concavity.

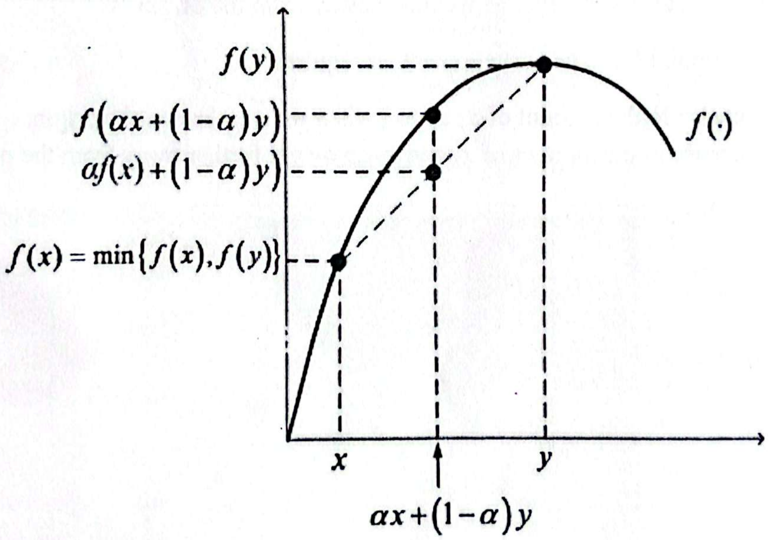


Figure: Concavity implies quasi-concavity

A concave $u(\cdot)$ exhibits diminishing marginal utility. That is, for an increase in the consumption bundle, the increase in utility is smaller as we move away from the origin.

The "jump" from one indifference curve to another requires:

- a slight increase in the amount of $x_1$and x_2 when we are close to the origin
- a large increase in the amount of x_1 and x_2 as we get further away from the origin

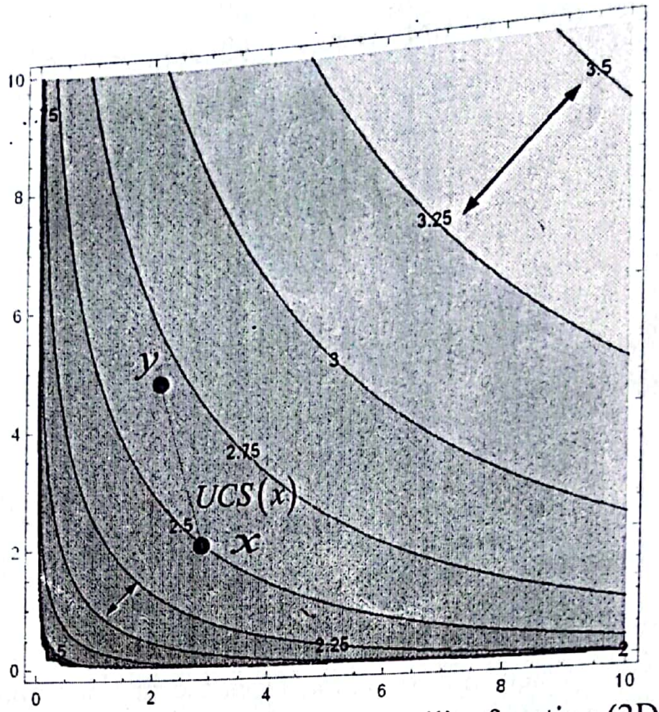


Fig: Concave and quasi-concave utility function (2D)

A convex $u(\cdot)$ exhibits increasing marginal utility. That is, for an increase in the consumption bundle, the increase in utility is larger as we move away from the origin.

The "jump" from one indifference curve to another requires:

- a large increase in the amount of x_1 and x_2 when we are close to the origin
- a small increase in the amount of x_1 and x_2 as we get further away from the origin

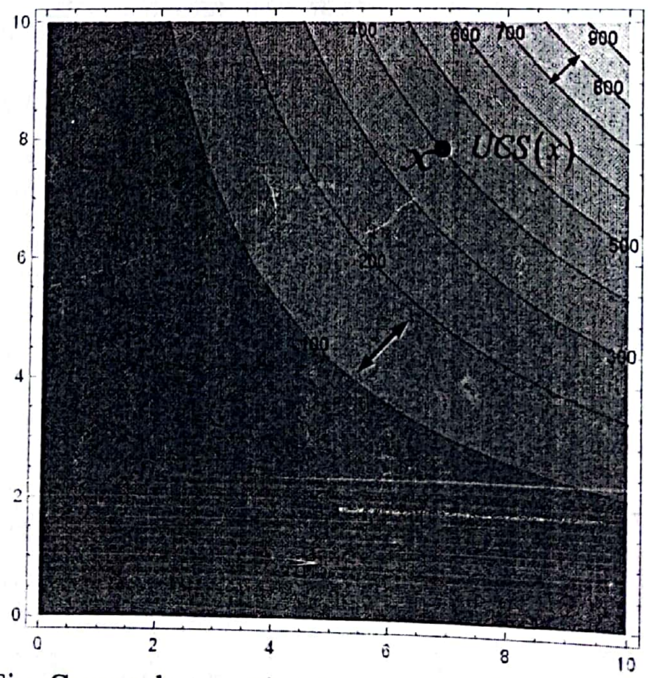


Fig: Convex but quasi-concave utility function (2D)

For any bundle x, the UCS(x) defines a convex set, and therefore the utility function satisfies the definition of quasi-concavity. Therefore, quasi-concavity (convex UCS) does not necessarily imply the concavity of a utility function.

In case of well-defined utility function, concavity implies quasi-concavity but quasi-concavity does not imply concavity.

Question 10: What are the typical utility functions used in Economic applications. Write down their functional form and calculate and interpret the notion of MRS.

Answer:
Here are some of the typical utility functions used in economic applications and their functional form:

i. Cobb-Douglas Utility Functions
ii. Perfect Substitutes
iii. Perfect Complements
iv. CES (constant elasticity of substitution) Utility Function
v. Quasi-linear Utility Function

i. Cobb-Douglas Utility Functions: In the case of two goods, x_1 and x_2.

$$u(x_1, x_2) = Ax_1^{\alpha}x_2^{\beta}$$

where $A, \alpha, \beta > 0$.

Applying logs on both sides, we get:

$$\log u = \log A + \alpha \log x_1 + \beta \log x_2$$

Hence, the exponents in the original $u(\cdot)$ can be interpreted as elasticities:

$$\epsilon_{u,x_1} = \frac{\partial u(x_1, x_2)}{\partial x_1} \cdot \frac{x_1}{u(x_1, x_2)} = \alpha A x_1^{\alpha-1} x_2^{\beta} \cdot \frac{x_1}{A x_1^{\alpha} x_2^{\beta}} = \alpha$$

Intuitively, a one-percent increase in the amount of good x_1 increases individual utility by α percent. A similar interpretation applies to $\epsilon_{u,x_2} = \beta$.

Special cases are those in which

- $\alpha + \beta = 1$, such as $u(x_1, x_2) = Ax_1^{\alpha}x_2^{1-\alpha}$
- $A = 1$, such as $u(x_1, x_2) = x_1^{\alpha}x_2^{\beta}$
- $A = \alpha = \beta = 1$, such as $u(x_1, x_2) = x_1 x_2$

Besides yielding positive marginal utilities for all goods, whereby $\frac{\partial u}{\partial x_1} > 0$ and $\frac{\partial u}{\partial x_2} > 0$, and thus satisfying strong monotonicity, the Cobb-Douglas utility function yields a diminishing MRS, since

$$MRS_{x_1,x_2} = \frac{\alpha A x_1^{\alpha-1} x_2^{\beta}}{\beta A x_1^{\alpha} x_2^{\beta-1}} = \frac{\alpha x_2}{\beta x_1}$$

which is decreasing in x_1. Hence, indifference curves become flatter as x_1 increases.

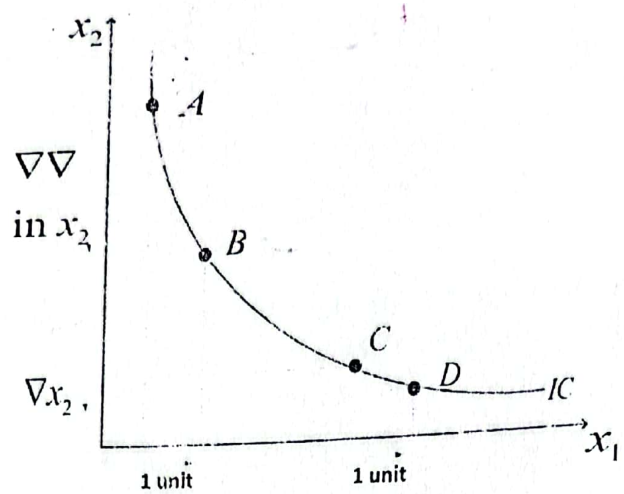


Fig: Cobb-Douglas preference

Individuals with Cobb–Douglas preferences are willing to give up many units of x_2 in order to increase their consumption of x_1 by one unit when they do not enjoy many units of x_1 (movement from point A to B in the figure), but such willingness to give up units of x_2 to obtain one more unit of x_1 decreases as x_1 becomes relatively abundant (movement from point C to D).

ii. **Perfect Substitutes:** In the case of two goods, x_1 and x_2,

$$u(x_1, x_2) = Ax_1 + Bx_2$$

where $A, \alpha, \beta > 0$.

As a consequence, the marginal utility of every good is constant, that is, $\frac{\partial u}{\partial x_1} = A$ for all x_1 and $\frac{\partial u}{\partial x_2} = B$ for all x_2. The MRS is also constant, since $MRS_{x_1,x_2} = \frac{A}{B}$ does not depend on the amount of goods 1 or 2. Therefore indifference curves are straight lines with a slope of $-\frac{A}{B}$, as shown in figure.

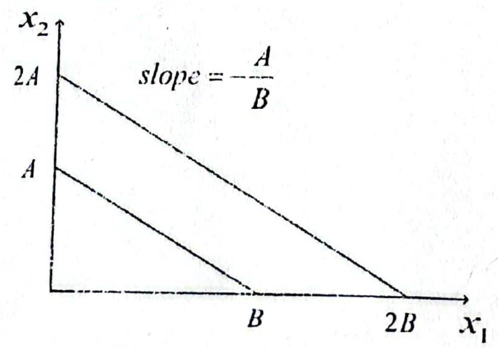


Fig: Perfect substitutes

Intuitively, the individual is willing to give up $\frac{A}{B}$ units of x_2 in order to obtain one more unit of x_1 and keep his utility level unaffected. Such willingness is, unlike in the Cobb–Douglas case, independent in the relative abundance of the two goods.

Examples: butter and margarine, coffee, and black tea, or two brands of unflavored mineral water.

iii. **Perfect Complements:** In the case of two goods, x_1 and x_2.

$$u(x_1, x_2) = A \cdot \min\{\alpha x_1, \beta x_2\}$$

where $A, \alpha, \beta > 0$.

Intuitively, increasing one of the goods without increasing the amount of the other good entails no increase in utility, which only goes up if the amounts of both goods increase (although they do not need to increase by the same amount).

The indifference curve is right angle with a kink at $\alpha x_1 = \beta x_2$, which, after rearranging, yields ray $x_2 = \frac{\alpha x_1}{\beta}$ from the origin, as depicted in figure.

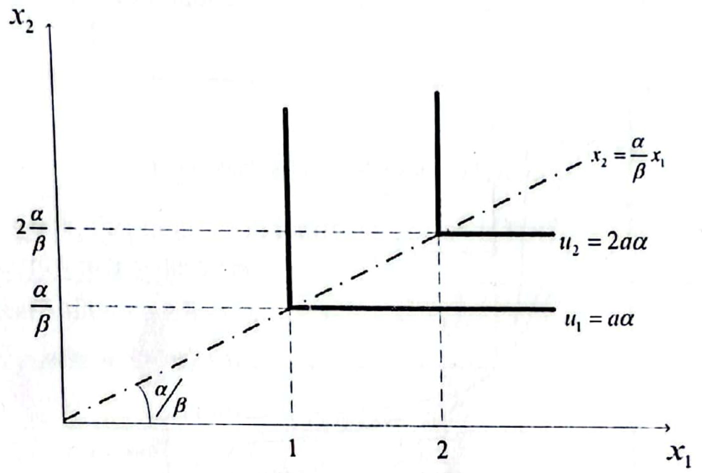


Fig: Perfect complements

The slope of such ray, $\frac{\alpha}{\beta}$, indicates the rate at which goods x_1 and x_2 must be consumed in order to achieve utility gains.

Special case: $\alpha = \beta$

$$u(x_1, x_2) = A \cdot \min\{\alpha x_1, \alpha x_2\}$$
$$= A\alpha \cdot \min\{x_1, x_2\}$$
$$= B \cdot \min\{x_1, x_2\} \; if \; B \equiv A\alpha$$

Examples: cars and gasoline, or peanut butter and jelly. Other food recipes, which often require the use of ingredients in a fixed proportion, are good examples as well.

iv. **Constant elasticity of substitution (CES)**: In the case of two goods, x_1 and x_2,

$$u(x_1, x_2) = \left[a x_1^{\sigma - \frac{1}{\sigma}} + b x_1^{\sigma - \frac{1}{\sigma}} \right]^{\frac{\sigma}{\sigma - 1}}$$

where σ measures the elasticity of substitution between goods x_1 and x_2.

In particular,

$$\sigma = \frac{\partial\left(\frac{x_2}{x_1}\right)}{\partial MRS_{1,2}} \frac{MRS_{1,2}}{\frac{x_2}{x_1}}$$

Here,

- σ = 0 under perfect complements (where substituting units of x_1 and x_2 without affecting the utility level is impossible),
- σ = +∞ under perfect substitutes (where goods x_1 and x_2 can be exchanged at the same rate),
- σ = 1 under the Cobb–Douglas utility function.

Figure depicts the indifference curves of the CES utility function evaluated at different values of σ.

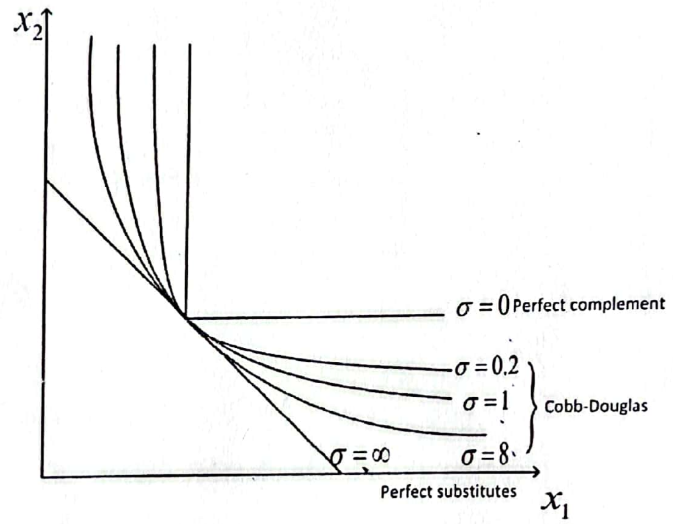


Fig: Constant elasticity of substitution (CES)

CES utility function is often presented as

$$u(x_1, x_2) = \left[ax_1^{\rho} + bx_1^{\rho}\right]^{\frac{1}{\rho}}$$

where, $\rho \equiv \frac{\sigma - 1}{\sigma}$.

v. **Quasi-linear Utility Function:** In the case of two goods, x_1 and x_2,

$$u(x_1, x_2) = v(x_1) + bx_2$$

where x_2 enters linearly, b > 0, and $v(x_1)$ is a nonlinear function of x_1. For example, $v(x_1) = a \ln x_1$ or $v(x_1) = ax_1^{\alpha}$, where $a > 0$ and $a \neq 1$.

The MRS is constant in the good that enters linearly in the utility function (x_2 in our case). In other words, for a given level of good 1, an increase in the amounts of good 2 (the desirable good that enters linearly) does not affect the slope of the indifference curve as depicted in figure.

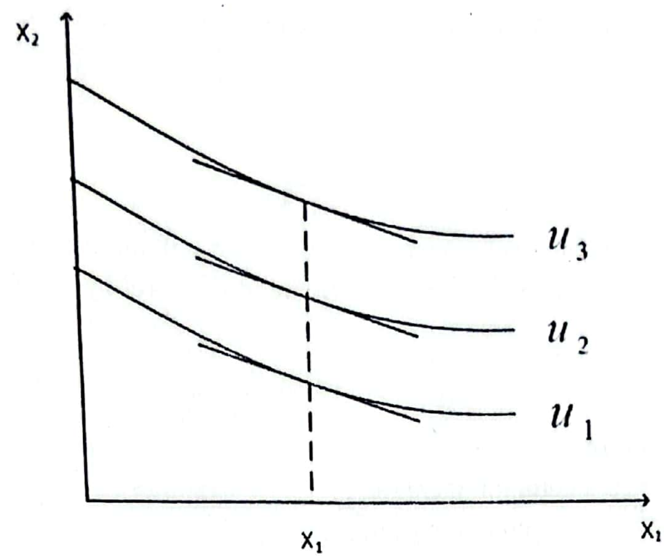


Fig: MRS of quasilinear preferences

Hence, at a given level of x_1, a move to indifference curves farther away from the origin (larger amount of x_2) does not change its slope.

For the quasi-linear utility function $u(x_1, x_2) = v(x_1) + bx_2$, we have that $MU_{x_1} = \boxed{\frac{\partial v}{\partial x_1}}$ and $MU_{x_2} = b$, which implies that

$$MRS_{x_1, x_2} = \frac{\frac{\partial v}{\partial x_1}}{b}$$

which is constant in the good entering linearly, x_2.

Quasilinear preferences are often used to represent the consumption of goods that are relatively insensitive to income.

Examples: garlic, toothpaste, etc.

Question 11: Examine the following properties of preference relations:

a) Homogeneity
b) Homotheticity

Answer:

Homogeneity: A utility function is homogeneous of degree k if varying the amounts of all goods by a common factor $\alpha > 0$ produces an increase in the utility level by α^k. That is, for the case of two goods, x_1 and x_2,

$$u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$$

where $\alpha > 0$. This allows for:

- $\alpha > 1$ in the case of a common increase
- $0 < \alpha < 1$ in the case of a common decrease

Properties of homogeneity: There are three properties of homogeneity.

i. The first-order derivative of a function $u(x_1, x_2)$, which is homogeneous of degree k, is homogeneous of degree $k - 1$. In particular, by homogeneity of degree k, $u(\alpha x_1, \alpha x_2) = \alpha^k u(x_1, x_2)$. Differentiating with respect to x_i for i = {1, 2} yields

$$\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_i} . \alpha = \alpha^k . \frac{\partial u(x_1, x_2)}{\partial x_i}$$

$$\Rightarrow u'(\alpha x_1, \alpha x_2) = \alpha^{k-1} u'(x_1, x_2)$$

Therefore $u'(x_1, x_2)$ is homogeneous of degree k − 1.

ii. The indifference curves of homogeneous functions are radial expansions of one another. That is, if two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve, i.e., $u(\alpha y) = u(\alpha z)$.

iii. The MRS of a homogeneous function is constant for all points along each ray from the origin. That is, the slope of the indifference curve at point y coincides with the slope at a "scaled-up version" of point y, αy, where $\alpha > 1$. Since the MRS at bundle $x = x_1, x_2$ is

$$MRS_{1,2}(x_1, x_2) = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial (x_1, x_2)}{\partial x_2}}$$

the MRS at $(\alpha x_1, \alpha x_2)$ is

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial (\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\alpha^{k-1} \frac{\partial (x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \frac{\partial (x_1, x_2)}{\partial x_2}} = \frac{\frac{\partial (x_1, x_2)}{\partial x_1}}{\frac{\partial (x_1, x_2)}{\partial x_2}}$$

where the last equality uses the first property. Hence, the MRS is unaffected along all the points crossed by a ray from the origin.

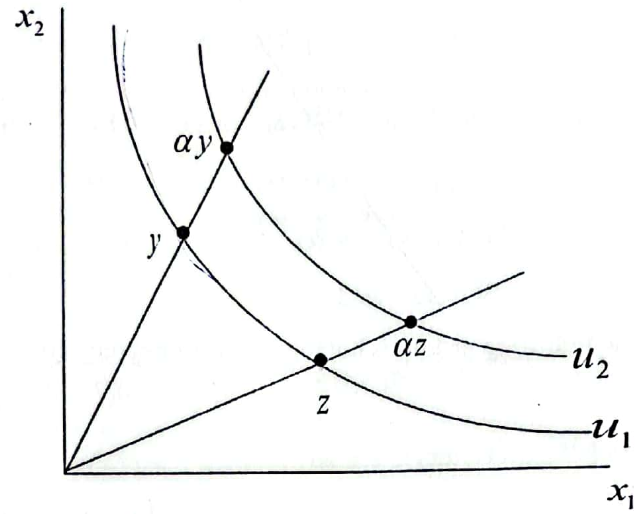


Fig: Homogeneous preference

Homotheticity: A utility function $u(x)$ is homothetic if it is a monotonic transformation of a homogeneous function. That is, $u(x) = g(v(x))$, where

- $g: \mathbb{R} \rightarrow \mathbb{R}$ is a strictly function, and
- $v: \mathbb{R}^n \rightarrow \mathbb{R}$ is homogeneous of degree k.

Properties of homotheticity: There are two properties of homotheticity.

i. If preferences, $u(x)$, is homothetic, and two bundles y and z lie on the same indifference curve, i.e., $u(y) = u(z)$, bundles αy and αz also lie on the same indifference curve, i.e., $u(\alpha y) = u(\alpha z)$ for all $\alpha > 0$, as shown in figure.

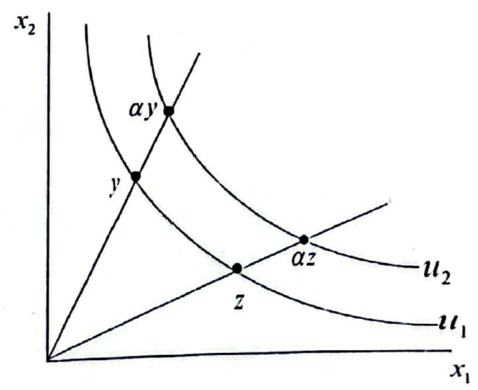


Fig: Homogeneous preference

Proof

If $u(y) = u(z)$ for two bundles y, z, then

$$u(\alpha y) = g(v(\alpha y)) = g(\alpha^k v(y))$$

where the first equality follows from homotheticity, and the second from the property that v(x) is homogeneous of degree k. Since we can construct a similar equality for bundle z,

$$u(\alpha z) = g(v(\alpha z)) = g(\alpha^k v(z))$$

and v(y) = v(z), implying that u(αy) = u(αz).

ii. The MRS of a homothetic function is homogeneous of degree zero.

Proof

Let us first find $MRS_{1,2}(\alpha x_1, \alpha x_2)$,

$$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial g}{\partial v} \cdot \frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}}$$

where, $u(x_1, x_2) \equiv g(v(x_1, x_2))$.

Canceling the $\frac{\partial g}{\partial v}$ terms yields

$$\frac{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial v(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\alpha^{k-1} \cdot \frac{\partial v(x_1, x_2)}{\partial x_1}}{\alpha^{k-1} \cdot \frac{\partial v(x_1, x_2)}{\partial x_2}}$$

Canceling the α^{k-1} terms yields

$$\frac{\frac{\partial v(x_1, x_2)}{\partial x_1}}{\frac{\partial v(x_1, x_2)}{\partial x_2}}$$

In summary,

✓ $$MRS_{1,2}(\alpha x_1, \alpha x_2) = \frac{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_1}}{\frac{\partial u(\alpha x_1, \alpha x_2)}{\partial x_2}} = \frac{\frac{\partial u(x_1, x_2)}{\partial x_1}}{\frac{\partial u(x_1, x_2)}{\partial x_2}} = \frac{\frac{\partial v(x_1, x_2)}{\partial x_1}}{\frac{\partial v(x_1, x_2)}{\partial x_2}} = MRS_{1,2}(x_1, x_2)$$

12. Proof that If a preference relation satisfies monotonicity and continuity, then there exists a utility function $u(\cdot)$ representing such preference relation.

Let us take a bundle $x \neq 0$

By monotonicity, $x \succsim 0$, where $0 = (0, 0, \dots, 0)$. Indeed, if bundle $x \neq 0$, then it contains positive amounts of at least one good, and, by monotonicity it is preferred to bundle 0.

Let $m \equiv \max\{x_1, x_2, \dots, x_n\}$ be the number of units of the most abundant good in bundle x. Let us define bundle M as the bundle where all components coincide with the highest component of bundle x. That is, $M \equiv (m, m, \dots, m)$

Bundle M hence increases the amounts of all goods in bundle except for the most abundant commodity, good 1 in the followin fig. And hence, by monotonicity we have that consumer prefers M to x, $M \succsim x$.

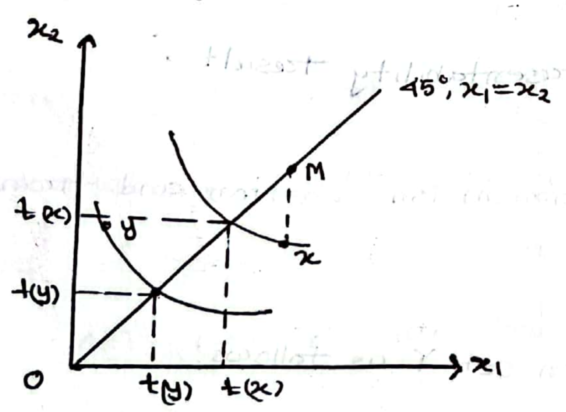


Fig: Existance of a utility function.

Bundles 0 and M are both on the main diagonal, since each of them contains the same amount of good x_1 and x_2. By continuity and monotonicity, there exists a bundle that is indifferent to x and lies on the main diagonal. By monotonicity, this bundle is unique.

we denote this bundle as $(t(x), t(x), \ldots, t(x))$. For completeness,

let us define $u(x) = t(x)$, which is a real number.

Applying the same steps for another bundle $y \neq x$, we obtain a similar bundle on the same diagonal, $(t(y), t(y), \ldots, t(y))$, which lies on the same indifference curve as bundle y. For completeness,

let $u(y) = t(y)$, which is also a real number. We know that,

$$x \sim ((t(x), t(x), \ldots, t(x))), \; y \sim (t(y), t(y), \ldots, t(y)) \text{ and,}$$

$$x \succsim y$$

Hence, by transitivity of preference relation $x \succsim y$, if and only if

$$x \sim (t(x), t(x), \ldots t(x)) \succsim (t(y), t(y), \ldots t(y)) \sim y$$

and by monotonicity, this is true as long as the real number $t(x)$ exceeds $t(y)$, meaning $t(x) \geq t(y)$,

$$x \succsim y \Leftrightarrow t(x) \geq t(y) \Leftrightarrow u(x) \geq u(y)$$

as required for our representability result.

18. Define production vector/transformation function and transformation frontier.

If we consider a production set, Y as follows:

$$Y = \{y \in \mathbb{R}^L : F(y) \leq 0\}$$

here, $F(y)$ is the transformation function. This function can be intuitively understood as a production function as following fig illustrates:

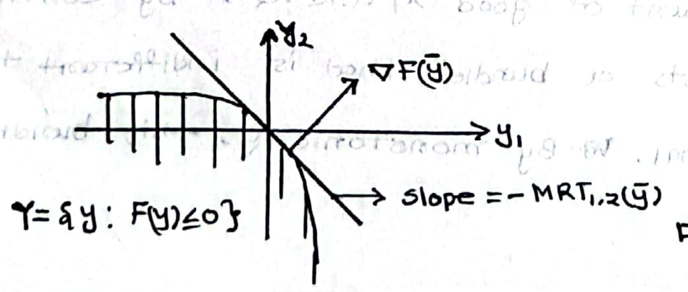


Fig: Production set and transformation frontier.

Here, the boundary of the production function indicates production plans for which $F(y)=0$. This boundary is the transformation frontier. Points strictly below the transformation frontier indicate (strictly) feasible production plans since $F(y)<0$. Therefore we can totally differentiate $F(y)$ as follows:

$$dF(y) = \frac{\delta F(y)}{\delta y_k} dy_k + \frac{\delta F(y)}{\delta y_k} \delta y_l$$

Now, we evaluate the expression above at any point $y=\bar{y}$ on the transformation frontier where $dF(\bar{y})=0$ holds. Solving for $\delta y_l / \delta y_k$, we obtain

$$\frac{\delta y_l}{\delta y_k} = -\frac{\delta F(\bar{y})/\delta y_k}{\delta F(\bar{y})/\delta y_l}$$

Where the term on R.H.S is $\dfrac{\delta F(\bar{y})/\delta y_k}{\delta F(\bar{y})/\delta y_l} = MRT_{l,k}(\bar{y})$

The marginal rate of transformation between good l and k, evaluated at $\bar{y}$, $MRT_{l,k}(\bar{y})$, measures how much the (net) output k can increase if the firm decreases the (net) output of good l by one unit, where the reduction in net output l implies a larger input usage. For instance, if good l denotes an input while k represents an output, $MRT_{l,k}(\bar{y})$ captures the increase in output y_k that the firm obtains from marginally increasing the amount of input y_l, it employs.

Q. Define production set. Explain the properties of production set.

Let us define a production plan $y=(y_1, y_2, \ldots, y_L)$ as a vector with L components. If a particular component of the vector is positive, it denotes that the firm produces positive amounts of good; for instance $y_2 > 0$. If instead a component of the y vector is negative, it denotes that the firm uses good 2 as an input in the production process. ~~A~~ production plans that are technologically feasible represented as a part of production set Y, as follows:

$$Y = \{ y \in R^L : F(y) \leq 0 \} \text{ ; where } F(y) \text{ is the transformation function.}$$

Different properties of production sets are given below:

1. Y is nonempty: That is, we have inputs and/or outputs. ~~For~~ in If a firm, despite using large quantities of inputs produces no outputs our production set will still be nonempty. This case is illustrated in following fig where the production set Y is nonempty, despite producing no units of outputs.

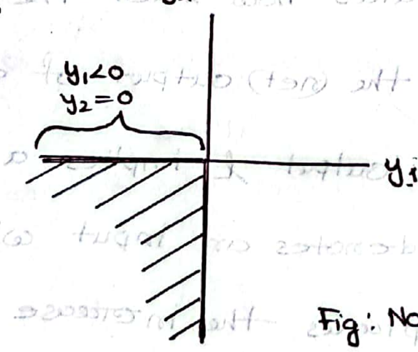


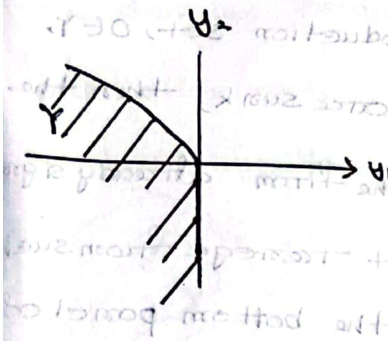
Fig: Non-empty production set Y

2. Y is closed: Hence the production set Y includes its boundary points.

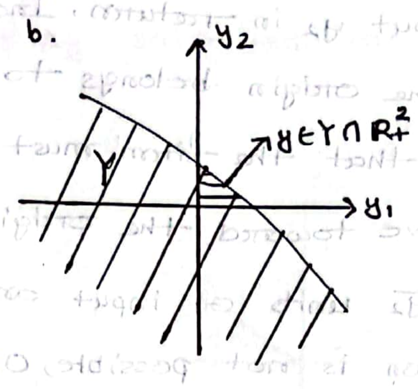
3. No free lunch: This property states that firm must use inputs in order to produce output. The top panel of the following fig represents a production set that satisfies no free lunch property, since the firm is using amounts of input y_1 in order to produce positive

amounts of output y_2. In contrast, the bottom panels of the fig illustrate production plans that violate the "no free lunch" property.

a.



b.



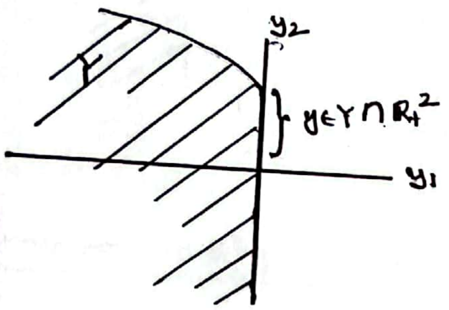


Fig: No free lunch property.

4. Possibility of inaction: This property states that firm can choose to be inactive, using no inputs and obtaining no output as a consequence. ~~At first~~ In other words, the input-output vector 0 is part of the production set. At first glance, a firm's possibility of inaction may seem applicable to many production processes, such as ~~the production set~~ depicted in the following fig.

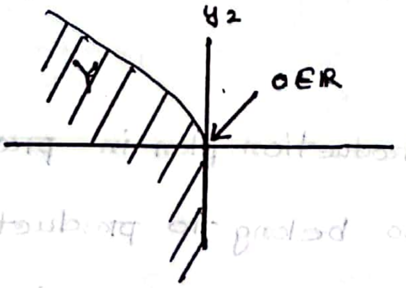
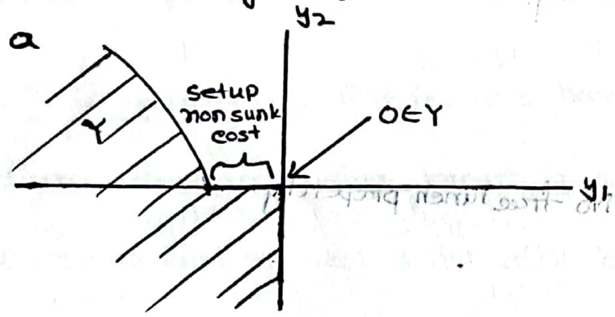


Fig: Possibility of inaction - I //

However, when firms face sunk costs, this property might be violated, meaning $0 \notin Y$. In particular, if the firm needs to use experiences a setup cost (fixed) as the top panel of the following fig depicts, the firm needs to use an amount of input y_1 without obtaining any output y_2 in ~~return~~. Inaction, however, is still possible since the origin belongs to the production set, $0 \in Y$. If the fixed costs that the firm must incur are sunk, then the firm cannot move toward the origin. If the firm already signed the purchase of $\bar{y}_2$ units of input and cannot renege from such a contract, inaction is not possible, $0 \notin Y$, as the bottom panel of the following fig illustrates.



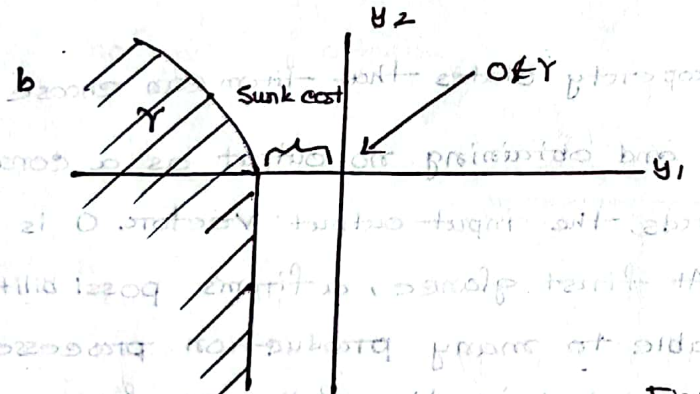


Fig: Possibility of inaction - II

5. <u>Free disposal</u>: If y is a production plan in production set Y, $y \in Y$, and $y' \leq y$, then y' must also belong to production set Y. Intuitively production plan y' is less efficient than production plan y. The production plan y' must must also belong to the firm's production set. Hence the producer can use more inputs without

he need to reduce its output, as he can despose of the additional inputs he doesnot need at no cost. The following fig illustrates the free disposal property being satisfied for two very different production sets.

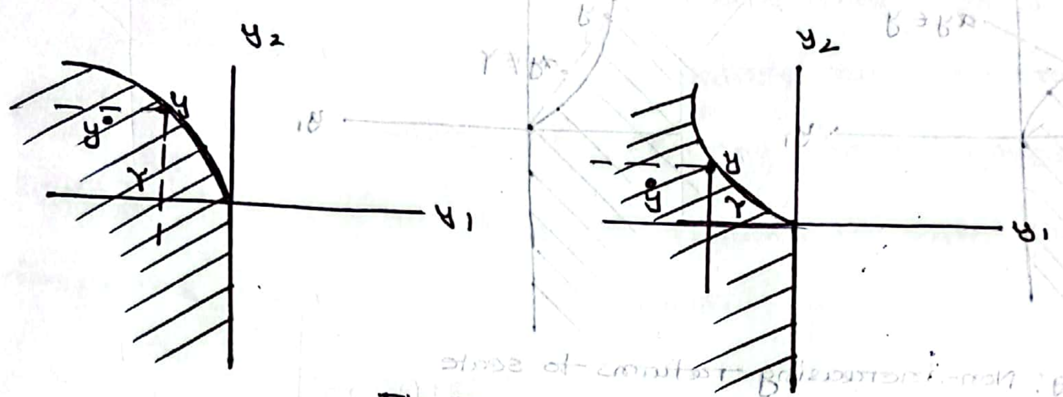
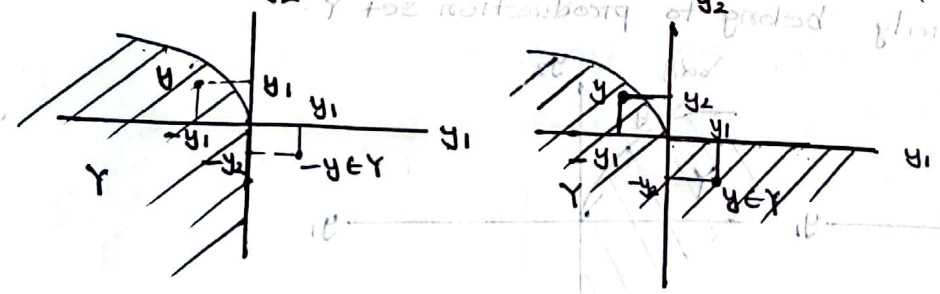


Fig: Free disposal

6. Irreversibility: Let us consider a production plan $y \in Y$, where y doesnot coincide with the origin, that is, $y \neq 0$. Then, production plan $-y$ cannot belong to Y. The following fig illustrates the no irreversibility property. This property highlights that there is no way back for this firm, in sense of transforming all its output into the exact amount of inputs that generated such output.

a. b.



7. Non-increasing returns to scale: Let us take a production plan $y \in Y$. Then any scaling down of this production plan $y, \alpha y$, where $\alpha \in [0,1]$ is also a part of production set. The following fig illustrates a production

set satisfying nonincreasing returns to scale in the left panel and a production set that violates non-increasing returns to scale on the right panel.

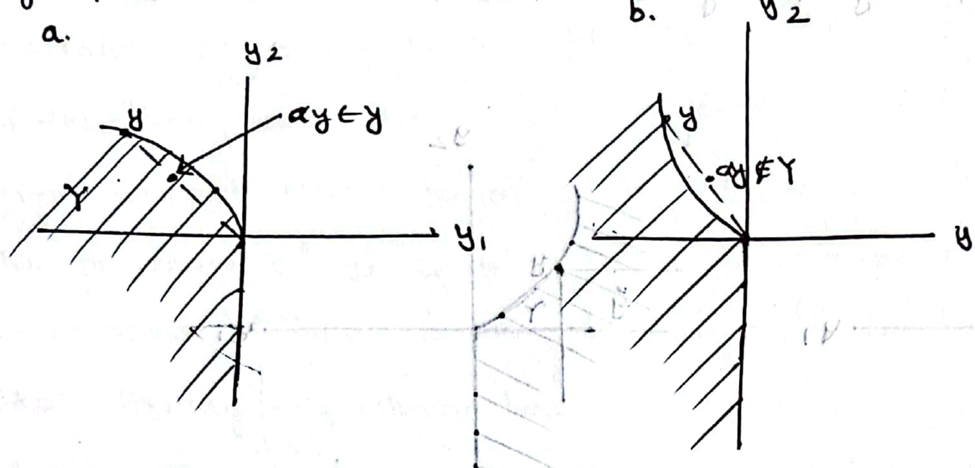


Fig: Non-increasing returns to scale

8. <u>Non-decreasing returns to scale:</u> Let us take a production plan $y \in Y$. Then any scaling up of this production plan αy, where $\alpha \geq 1$, is also part of production set Y. The following fig (left hand panel) depicts a production set satisfying non-decreasing returns to scale, since scaling up any production plan y yields a new production plan αy that also belongs to Y. In contrast, the right hand panel shows a production set that violates non-decreasing returns to scale: scaling up production plan y yields a new production plan αy that doesn't necessarily belong to production set Y.

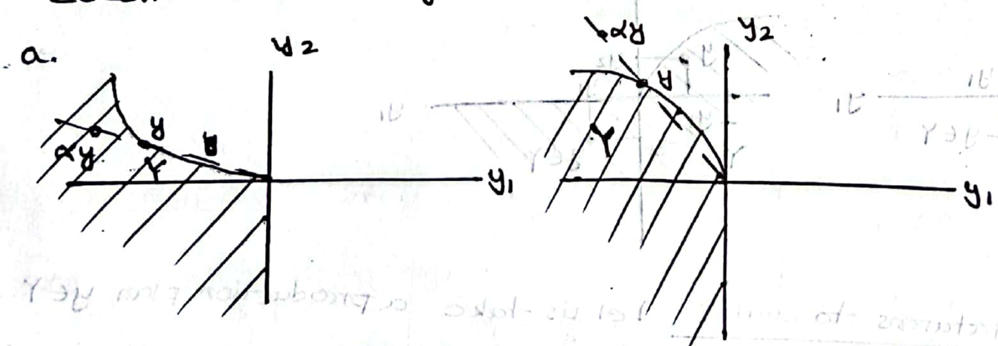


Fig: Non-decreasing returns to scale.

9. <u>Constant returns to scale:</u> Let us take a production plan $y \in Y$. Then production plan αy also belongs to production set Y, for any $\alpha \geq 0$. Hence both scaling down an original production plan y (when $\alpha \in [0,1]$) and scaling it up (when $\alpha > 1$) yields a new production plan that still belongs to production set Y. to satisfy constant ~~returns to scale~~

This point is illustrated in following fig: Graphically, for a production set Y to satisfy constant returns to scale, we need the transformation frontier to be straight line.

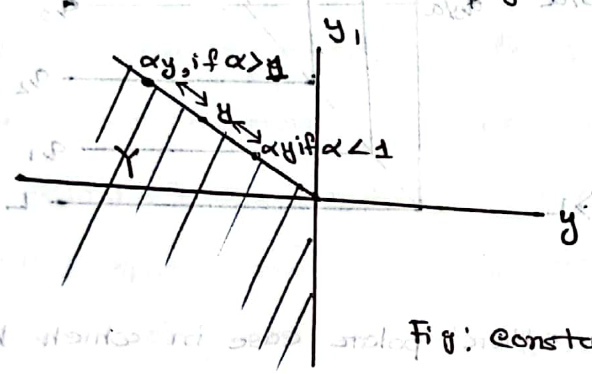


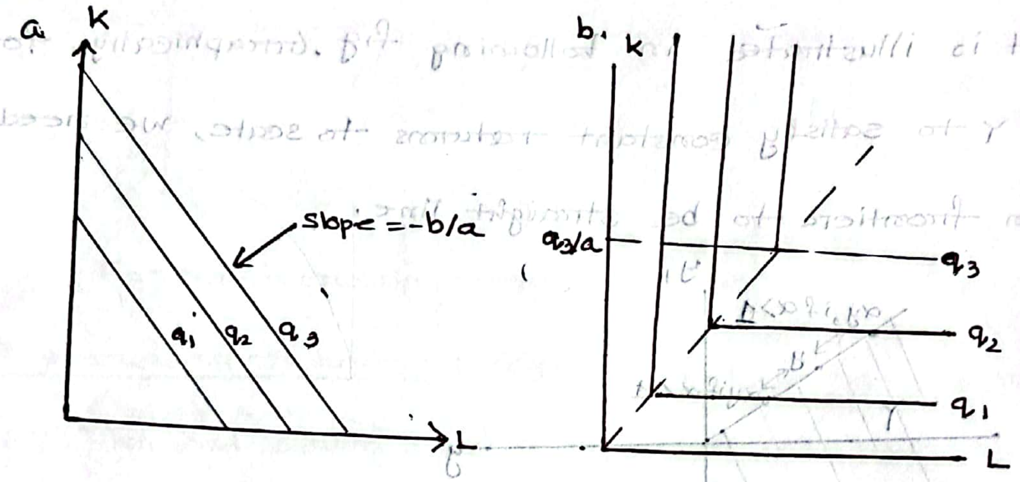
Fig: constant returns to scale.

Obtain the elasticity of substitution (EoS). Find the Elasticities of Substitution for the production function $q = f(k, L) = ak + bL$. What will be the EOS of the production function in fixed proportion.

The elasticity of substitution measures the proportionate change in the k/L ratio relative to the proportionate change in the $MRTS_{L,k}$ along an isoquant. That is, $\sigma = \left(\% \Delta \frac{k}{L}\right) / \left(\% \Delta MRTS_{Lk}\right)$

Let us consider a linear production function $q = f(k, L) = ak + bL$. Solving for k in production function $q = ak + bl$ yields an isoquant $k = q/a - (b/a)/l$, with slope $-b/a$, which is constant in labor.

Panel a of the following fig illustrates a linear production function in which labor and capital are perfect substitutes in production process. It implies that $MRTS_{LK}$ is constant in k/L and hence the elasticity of substitution yields an infinitely large number $\sigma = \infty$.



Let us next examine the other polar case in which both inputs must be used in fixed proportion, as $q = \min\{ak, bL\}$, where $a, b > 0$. In this case the $MRTS_{LK}$ changes dramatically from infinite (for labor amounts below the kink in the isoquants as depicted in panel b) to zero. This implies the change in $MRTS_{L,K}$ is infinite, yielding as a consequence a zero elasticity of substitution, namely $\sigma = 0$.

21. Define profit function. If the production set Y is closed and have fixed disposal, then what ~~responsibilities~~ properties do the profit function satisfies?

If we consider a profit maximization problem, $\max p.y$ subject to $y \in Y$, the value function resulting from this profit maximization problem, $\pi(P)$, denoted as the "profit function" of the firm, describes for every price p the highest amount of profits, chosen by the profit-maximizing production plan $y(P)$. More formally, we can define profit function as :

$$Pi(P) = P \cdot y(P);$$

that is the firms profit evaluated at $y = y(P)$.

Now, let us describe some properties of the profit function $\pi(P)$.

Let us assume that the production set Y is closed and satisfies the free disposal property :

1. **Homogeneity:** The profit function $\pi(P)$ is homogeneous of degree one in prices. That is increasing the prices of all inputs and outputs by the same factor λ produces a proportional increase in the firms profits, $\pi(\lambda P) = \lambda \pi(P)$. The profit function can be expressed as $\pi(P) = p.q - w.z$, where all L inputs (invector z) and the output q are evaluated at their profit-maximizing amounts. Scaling all prices by common factor λ, we obtain :

$$\lambda p.q - \lambda w.z = \lambda (p.q - w.z) = \lambda \cdot \pi(P)$$

which shows that the firm's profit increases in the same proportion (λ) as input and output prices, thus implying that homogeneity of degree one.

2. **Convex in price!** The profit function $\pi(P)$ is convex. The following fig describes this property. For a price vector p, the profit-maximizing production plan is $y(P)$, yielding an associated profit function $\pi(P)$. Similarly, for a price vector p', the profit maximizing production plan is $y(P)$, yielding an associated profit function $\pi(P)$

//

3. <u>Alternative representation:</u> If a production set Y is convex, then

$$Y = \{y \in R^L : py \leq \pi(p) \text{ for all } p >> 0\}$$

Intuitively, the production set Y can be alternatively represented by this "dual" set. It specifies that, for any given price vector p, all production vector y generate less profit $p.y$ than the profit maximizing production plan $y(p)$, which entails a profit function $\pi(p) = py(p)$. A graphical representation of this property is shown in following fig:

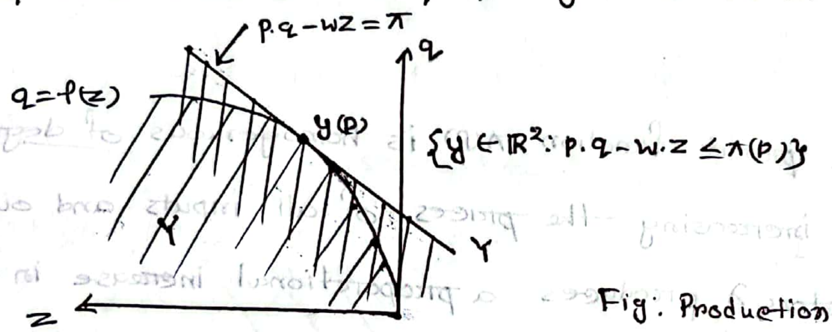


Fig: Production set Y is convex.

As the fig shows, in a convex production set Y, the supply corresponden $y(p)$ maximizes profits and the associated isoprofit line is $\pi = pq - wz$. Since all production plans (z, q) below this isoprofit line ~~is $\pi = pq$~~ yield a lower profit level for the firm $pq - wz \leq \pi(p)$. Alternatively, the isoprofit line $\pi = pq - wz$ can be represented by

$$q = \frac{\pi}{p} - \frac{w}{p} z$$

If the price vector w/p is constant, meaning input (output) prices are unaffected by different level of input usage, then the slope of the isoprofit line is constant in z. Therefore the set of (z, q) pair below profit function $\pi(p)$ define a convex set.

<u>22.</u> Consider a consumer with the utility function $u(x_1, x_2) = x_1^{1/4} x_2^{1/2}$

i) Derive the walrasian and compensated demand function.

ii) Consider $\{p^0, m^0\} = \{4, 4, 300\}$ a) compute the Indirect utilities and $\{p^1, m^0\} = \{2, 4, 300\}$ cost

iii) Calculate EV, CV (& ES) and examine their relationship and comment on the nature of the commodities.

i) Given, Cobb-Douglas utility function, $u(x_1, x_2) = x_1^{\frac{1}{4}} \cdot x_2^{\frac{1}{2}}$

Budget constraint, $m = P_1 x_1 + P_2 x_2$

Walrasian / Marshallian demand function → Maximize utility

$$x_1(P, W) = \frac{\alpha W}{(\alpha+\beta)P_1} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{2}} \cdot \frac{m}{P_1} = \frac{\frac{1}{4}}{\frac{3}{4}} \frac{m}{P_1} = \frac{1}{3}\frac{m}{P_1}$$

$$x_2(P, W) = \frac{\beta W}{(\alpha+\beta)P_2} = \frac{\frac{1}{2}}{\frac{1}{4}+\frac{1}{2}} \cdot \frac{m}{P_2} = \frac{\frac{1}{2}}{\frac{3}{4}} \frac{m}{P_2} = \frac{2}{3}\frac{m}{P_2}$$

Compensated / Hicksian demand function → Minimize expenditure

$$h_1(P, u) = \left(\frac{\alpha}{\beta} \cdot \frac{P_2}{P_1}\right)^{\frac{\beta}{\alpha+\beta}} \cdot u^{\frac{1}{\alpha+\beta}} = \left(\frac{\frac{1}{4}}{\frac{1}{2}} \cdot \frac{P_2}{P_1}\right)^{\frac{\frac{1}{2}}{\frac{1}{4}+\frac{1}{2}}} \cdot u^{\frac{1}{\frac{1}{4}+\frac{1}{2}}} = \left(\frac{P_2}{2P_1}\right)^{\frac{2}{3}} \cdot u^{\frac{4}{3}}$$

$$h_2(P, u) = \left(\frac{\beta}{\alpha} \cdot \frac{P_1}{P_2}\right)^{\frac{\alpha}{\alpha+\beta}} u^{\frac{1}{\alpha+\beta}} = \left(\frac{\frac{1}{2}}{\frac{1}{4}} \cdot \frac{P_1}{P_2}\right)^{\frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{2}}} \cdot u^{\frac{1}{1/4+1/2}} = \left(\frac{2P_1}{P_2}\right)^{1/3} u^{\frac{4}{3}}$$

ii) Indirect Utility:

$$\text{Indirect utility } V = v(P, W) = \left(\frac{\alpha}{\alpha+\beta} \frac{W}{P_1}\right)^{\alpha} \left(\frac{\beta}{\alpha+\beta} \frac{W}{P_2}\right)^{\beta} = \left(\frac{\alpha}{P_1}\right)^{\alpha} \left(\frac{\beta}{P_2}\right)^{\beta} \left(\frac{W}{\alpha+\beta}\right)^{\alpha+\beta}$$

Indirect utility at initial price $u^0 = \left(\frac{1/4}{4}\right)^{\frac{1}{4}} \left(\frac{\frac{1}{2}}{4}\right)^{\frac{1}{2}} \left(\frac{300}{1/4+1/2}\right)^{1/4+1/2} = 15.81$

Indirect " " final " $U^1 = \left(\frac{\frac{1}{4}}{2}\right)^{1/4} \left(\frac{\frac{1}{2}}{4}\right)^{\frac{1}{2}} \left(\frac{300}{1/4+1/2}\right)^{1/4+1/2} = 18.80$

Costs/Expenditure:

$$\text{Cost} \to e(P, u) = P_1^{\frac{\alpha}{\alpha+\beta}} P_2^{\frac{\beta}{\alpha+\beta}} U^{\frac{1}{\alpha+\beta}} \left[\left(\frac{\alpha}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\beta}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}}\right]$$

Costs under initial prices, p^0 and maximal utility level $U^0 = V(P^0, W)$

$$e(P^0, u^0) = 4^{\frac{1}{1/4+1/2}} 4^{\frac{1/2}{1/4+1/2}} \cdot (15.81)^{\frac{1}{1/4+1/2}} \left[\left(\frac{1/4}{1/2}\right)^{\frac{1/2}{1/4+1/2}} + \left(\frac{1/2}{1/4}\right)^{\frac{1/4}{1/4+1/2}}\right]$$

$$= 4^{1/3}\, 4^{2/3}\, (15.81)^{4/3} \left[(1/2)^{2/3} + (2)^{1/3}\right]$$

$$= 299.96 \approx 300$$

Costs under initial prices, p° and maximal utility level, $U' \equiv V(P^{\circ}, w)$,

$$e(P^{\circ}, U^{1}) = 4^{1/3}\, 4^{\frac{2}{3}}\, (18.80)^{4/3} \left[\left(\tfrac{1}{2}\right)^{2/3} + (2)^{-1/3}\right] = 377.895$$

Costs under ~~initial~~ prices, p^{1} and maximal utility level, $u^{\circ} \equiv V(P^{\circ}, w)$,

$$e(P', u^{\circ}) = (2)^{1/3}\, 4^{2/3}\, (15.81)^{4/3} \left[\left(\tfrac{1}{2}\right)^{2/3} + (2)^{1/3}\right] = 238.08$$

Costs under new prices P^{1} and maximal utility level, $u^{1} \equiv V(P^{1}, w)$

$$e(P', u^{1}) = 2^{1/3}\, 4^{2/3}\, (18.80)^{4/3} \left[\left(\tfrac{1}{2}\right)^{2/3} + (2)^{1/3}\right] = 299.94 \approx 300$$

iii) Equivalent variation: (EV)

The amount of money that the consumer would be willing to give up before the price increase to be as well off as after the price increase.

$$EV(P^{\circ}, P^{1}, w) = e(P^{\circ}, U^{1}) - e(P^{\circ}, U^{\circ}) = 377.895 - 300 = 77.9 \approx 78$$

Compensated variation (CV):

The amount of money that a consumer would need after a price increase to be as well off as before the price increase.

$$CV(P^{\circ}, P^{1}, w) = e(P', U') - e(P', U^{\circ}) = 300 - 238.08 = 61.92 \approx 62$$

Consumer surplus: (CS)

Price of commodity 1 decreased by 50%, from $P^{\circ} = 4$ to $P_1' = 2$.

The change in consumer surplus is:

$$es = \int_2^4 x_1 (P, w)\, dP_1 \quad \Big| \; x_1(P, w) = \frac{1}{3}\frac{m}{P_1} \longrightarrow \text{Watrasian demand.}$$

$$= \int_2^4 \left(\frac{1}{3} \frac{m}{P_1}\right) dP_1$$

$$= \int_2^4 \left(\frac{1}{3} \times \frac{300}{P_1}\right) dP_1$$

$$= 100 \int_2^4 \frac{1}{P_1}\, dP_1$$

$$= 69.31$$

<u>Relationship between ~~es~~ EV, cv and es:</u>

$$EV > es > cv$$

<u>Nature of commodities:</u>

As $EV > es > cv$, for the decrease in price of commodity 1, the commodities are normal goods.

**

15. Considerr the cobb-douglas utility function:

$$U(x_1, x_2) = k\, x_1^{\alpha}\, x_2^{1-\alpha}$$

For some $\alpha \in (0,1)$ and $k > 0$

i) Derive the walrasian demand function and examine their properties.

ii) How do you get the indirect utility function? Explain their properties. Verify Roy's identity.

iii) Solve the expenditure maximizing problem and explain the property of the compensated demand function.

i) Here, $U(x_1, x_2) = k\, x_1^{\alpha}\, x_2^{1-\alpha}$

s.t $\quad p_1 x_1 + p_2 x_2 = \omega$

Setting up the lagrangian—

$$L(x_1, x_2, \lambda) = k x_1^{\alpha} x_2^{1-\alpha} + \lambda(\omega - p_1 x_1 - p_2 x_2)$$

FOC: $\dfrac{\delta L}{\delta x_1} = \alpha k x_1^{\alpha-1} x_2^{1-\alpha} - \lambda p_1 = 0 \quad \cdots (i)$

$\dfrac{\delta L}{\delta x_2} = \cancel{\alpha}\, (1-\alpha) k x_1^{\alpha} x_2^{-\alpha} - \lambda p_2 = 0 \quad \cdots (ii)$

$\dfrac{\delta L}{\delta \lambda} = \omega - p_1 x_1 - p_2 x_2 = 0$

From (i) $\alpha k x_1^{\alpha-1} x_2^{1-\alpha} = \lambda p_1 \cdots (iii)$

From (ii) $(1-\alpha) k x_1^{\alpha} x_2^{-\alpha} = \lambda p_2 \cdots (iv)$

(iii) ÷ (iv) $\Rightarrow \dfrac{\alpha k x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) k x_1^{\alpha} x_2^{-\alpha}} = \dfrac{\lambda p_1}{\lambda p_2}$

$\Rightarrow \dfrac{\alpha}{1-\alpha} \dfrac{\cancel{x_1^{\alpha}}}{\cancel{x_1^{\alpha}}} \dfrac{1}{x_1}\, x_2 \dfrac{\cancel{x_2^{-\alpha}}}{\cancel{x_2^{-\alpha}}} = \dfrac{p_1}{p_2}$

$\Rightarrow \dfrac{\alpha}{1-\alpha} \dfrac{x_2}{x_1} = \dfrac{p_1}{p_2}$

$\Rightarrow p_2 x_2 = \dfrac{1-\alpha}{\alpha} p_1 x_1$

ii) #

From the budget constraint -

$p_1x_1 + p_2x_2 = \omega$

$\Rightarrow p_1x_1 + \frac{1-\alpha}{\alpha} p_1x_1 = \omega$

$\Rightarrow p_1x_1 \left(1 + \frac{1-\alpha}{\alpha}\right) = \omega$

$\Rightarrow p_1x_1 \left(\frac{\alpha + 1 - \alpha}{\alpha}\right) = \omega$

$\Rightarrow p_1x_1 \frac{1}{\alpha} = \omega$

$\therefore x_1 = \frac{\alpha\omega}{p_1}$

Similarly, $x_2 = \frac{(1-\alpha)\omega}{p_2}$

$\therefore$ Walrasian demand for x_1, $x_1 = \frac{\alpha\omega}{p_1}$

" " " x_2, $x_2 = \frac{(1-\alpha)\omega}{p_2}$

Examining their properties:

① $x_1 = \frac{\alpha\omega}{p_1}$ or, $x_1(p,\omega) = \frac{\alpha\omega}{p_1}$

$x_1(\lambda p, \lambda\omega) = \frac{\alpha\lambda\omega}{\lambda p_1} = \frac{\alpha\omega}{p_1} = x_1(p,\omega)$ $[\lambda > 0]$

$x_2(p,\omega) = \frac{(1-\alpha)\omega}{p_2}$

$x_2(\lambda p, \lambda\omega) = \frac{(1-\alpha)\lambda\omega}{\lambda p_2} = \frac{(1-\alpha)\omega}{p_2} = x_2(p,\omega)$

So, it is homogeniety of degree zero.

② Walras's law $p\,x(p,\omega) = \omega$

Here, $x_1(p_1,\omega) = \frac{\alpha\omega}{p_1}$

$\Rightarrow p_1x_1(p_1,\omega) = \alpha\omega$

Similarly, $p_2x_2(p_2,\omega) = (1-\alpha)\omega$

The demand for every good k is increasing in wealth, ω, decreasing in its own price, p_i, but unaffected by the price of other good, p_l, $l \neq k$.

ii) Here, $x_1 = \frac{\alpha\omega}{P_1}$ and $x_2 = \frac{(1-\alpha)\omega}{P_2}$

So, the indirect utility function, $V(P,\omega) = \left(x_1(P,\omega)^{\alpha},\ x_2(P,\omega)^{1-\alpha}\right)$

$$= \left(\left(\frac{\alpha\omega}{P_1}\right)^{\alpha} \cdot \left(\frac{(1-\alpha)\omega}{P_2}\right)^{1-\alpha}\right)$$

$$= \frac{\alpha^{\alpha}\omega^{\alpha}}{P_1^{\alpha}} (1-\alpha)^{1-\alpha} \frac{\omega^{1-\alpha}}{P_2^{1-\alpha}}$$

$$\therefore\ V(P,\omega) = \left(\frac{\alpha}{P_1}\right)^{\alpha} \left(\frac{1-\alpha}{P_2}\right)^{1-\alpha} \cdot \omega$$

So, the indirect utility function is decreasing in the price of both goods, increasing in the consumer's wealth, ω. [Properties]

$$V(\lambda P, \lambda\omega) = \left(\frac{\alpha}{\lambda P_1}\right)^{\alpha} \left(\frac{1-\alpha}{\lambda P_2}\right)^{1-\alpha} \lambda\omega$$

$$= \left(\frac{\alpha}{P_1}\right)^{\alpha} \frac{1}{\lambda^{\alpha}} \left(\frac{1-\alpha}{P_2}\right)^{1-\alpha} \frac{1}{\lambda^{1-\alpha}} \cdot \lambda\omega$$

$$= \left(\frac{\alpha}{P_1}\right)^{\alpha} \left(\frac{1-\alpha}{P_2}\right)^{1-\alpha} \frac{1}{\lambda} \cdot \lambda\omega$$

$$= \left(\frac{\alpha}{P_1}\right)^{\alpha} \left(\frac{1-\alpha}{P_2}\right)^{1-\alpha} \cdot \omega = V(P,\omega)$$

So, it is homogeneous to degree zero.

The Roy's identity:

$$\frac{-\delta V(P,\omega)/\delta P_k}{\delta V(P,\omega)/\delta\omega} = x_k(P,\omega) \quad \text{For every good } k$$

Here, $\frac{\delta V(P,\omega)}{\delta P_1} = -\frac{\alpha}{P_1}\left(\frac{\alpha}{P_1}\right)^{\alpha}\left(\frac{1-\alpha}{P_2}\right)^{1-\alpha} \cdot \omega$

$\frac{\delta V(P,\omega)}{\delta\omega} = \left(\frac{\alpha}{P_1}\right)^{\alpha}\left(\frac{1-\alpha}{P_2}\right)^{1-\alpha}$

So, $\frac{-\delta V(P,\omega)/\delta P_1}{\delta P(P,\omega)/\delta\omega} = \frac{\alpha}{P_1}\omega = x_1(P,\omega)$

Similarly, $\frac{-\delta V/\delta P_2}{\delta V/\delta\omega} = \frac{1-\alpha}{P_2}\omega = x_2(P,\omega)$; Roy's identity varified.

$\therefore x_2 =$

iii) Given, $u(x_1,x_2) = k\, x_1^{\alpha}\, x_2^{1-\alpha}$

$\alpha \in (0,1)$ and $k>0$

s.t $p_1x_1 + p_2x_2 = \omega$

We can find the Hicksian demand (compensated demand) function for goods x_1 and x_2 by solving the EMP. Taking the FOC with respect to x_1, x_2 and μ, yields,

$$\min p_1x_1 + p_2x_2 \quad \text{s.t } k\, x_1^{\alpha} x_2^{1-\alpha} \geq \mu$$

[μ না u হবে]

$$\Rightarrow p_1 \geq \mu k \alpha x_1^{\alpha-1} x_2^{1-\alpha}$$

$$p_2 \geq \mu k(1-\alpha) x_1^{\alpha} x_2^{-\alpha} \quad \text{and}$$

$$k\, x_1^{\alpha} x_2^{1-\alpha} \geq u$$

Here, μ = lagrange multiplier of this minimization problem.

For interior solutions:

$$p_1 = \mu k \alpha x_1^{\alpha-1} x_2^{1-\alpha}$$

$$p_2 = \mu k(1-\alpha) x_1^{\alpha} x_2^{-\alpha}$$

$$k\, x_1^{\alpha} x_2^{1-\alpha} = u$$

So, $$\frac{p_1}{p_2} = \frac{\mu k}{\mu k} \frac{\alpha}{1-\alpha} \frac{x_1^{\alpha-1}}{x_1^{\alpha}} \frac{x_2^{1-\alpha}}{x_2^{-\alpha}}$$

$$\Rightarrow \frac{p_1}{p_2} = \frac{\alpha}{1-\alpha} \frac{x_2}{x_1}$$

$$k x_1^{\alpha} x_2^{1-\alpha} = u$$

$$\Rightarrow x_1^{\alpha} = \frac{u}{k} \frac{1}{x_2^{1-\alpha}}$$

$$\Rightarrow \left[\frac{\alpha}{1-\alpha} \frac{p_2}{p_1} . x_2\right]^{\alpha} = \frac{u}{k} \frac{1}{x_2^{1-\alpha}}$$

$$\Rightarrow \left(\frac{\alpha}{1-\alpha}\right)^{\alpha} . \left(\frac{p_2}{p_1}\right)^{\alpha} . x_2^{\alpha} = \frac{u}{k} \frac{x_2^{\alpha}}{x_2}$$

$$\Rightarrow \frac{u}{k} . \frac{1}{x_2} = \left(\frac{\alpha}{1-\alpha}\right)^{\alpha} \left(\frac{p_2}{p_1}\right)^{\alpha}$$

$$\Rightarrow \frac{u}{k} \frac{1}{\left(\frac{\alpha}{1-\alpha}\right)^{\alpha}} \frac{1}{\left(\frac{p_2}{p_1}\right)^{\alpha}} = x_2$$

$$\Rightarrow x_2 = \frac{u}{k}\left(\frac{1-\alpha}{\alpha}\right)^{\alpha} \left(\frac{p_1}{p_2}\right)^{\alpha}$$

$$\therefore x_2 = \frac{u}{k}\left(\frac{(1-\alpha)P_1}{\alpha P_2}\right)^{\alpha}$$

$$\text{So, } x_1 = \frac{u}{k}\left(\frac{\alpha P_2}{(1-\alpha)P_1}\right)^{1-\alpha}$$

This is Hicksian (Compensated) Demand function.

$$\text{So, } h_1(P,u) = \frac{u}{k}\left[\frac{\alpha P_2}{(1-\alpha)P_1}\right]^{1-\alpha}$$

$$h_2(P,u) = \frac{u}{k}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{\alpha}$$

$$\text{Now, } h_1(\lambda P, u) = \frac{u}{k}\left[\frac{\alpha \lambda P_2}{(1-\alpha)\lambda P_1}\right]^{1-\alpha}$$

$$= \frac{u}{k}\left[\frac{\alpha P_2}{(1-\alpha)P_1}\right]^{1-\alpha} = h_1(P,u)$$

Similarly, $h_2(\lambda P_2, u) = h_1(P_2, u)$

So, Hicksian demands satisfy homogeneity of degree zero in prices.

The Hicksian demands are also unique because, for a given price vectors, a given utility level u and a specific value for α $(1-\alpha)$, we can obtain a precise quantity demand for goods 1 and 2.

$$\text{Now, } u(h_1(P,u), h_2(P,u)) = k\,[h_1(P,u)]^{\alpha}\,[h_2(P,u)]^{1-\alpha}$$

$$= k\left(\frac{u}{k}\right)^{\alpha}\left[\frac{\alpha P_2}{(1-\alpha)P_1}\right]^{(1-\alpha)\alpha}\left(\frac{u}{k}\right)^{1-\alpha}\left[\frac{(1-\alpha)P_1}{\alpha P_2}\right]^{\alpha(1-\alpha)}$$

$$= k\,\frac{u^{\alpha}}{k^{\alpha}}\,\frac{u^{1-\alpha}}{k^{1-\alpha}}$$

$$= k\,\frac{u}{k} = u$$

So, the "no excess utility" property holds.

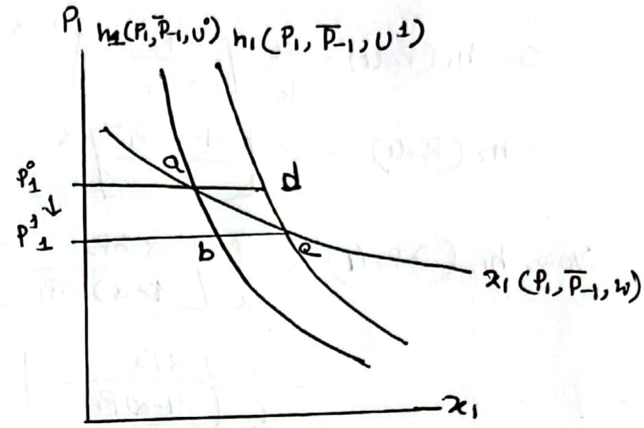
Graphically show that for a decrease in the price of a commodity (when both goods are normal), $EV > CS > CV$ and for price increase, $EV < CS < CV$

Price decrease:

$x_1(P_1, \bar{P}_{-1}, w)$ shows the walrasian demand curve.

$h_1(P_1, \bar{P}_{-1}, U^0)$ shows the hickshian demand curve at old utility level

$h_1(P_1, \bar{P}_{-1}, U^1)$ shows . . . . new . . . . after price decrease.



$EV = P_1^0 P_1^1 ed$ area

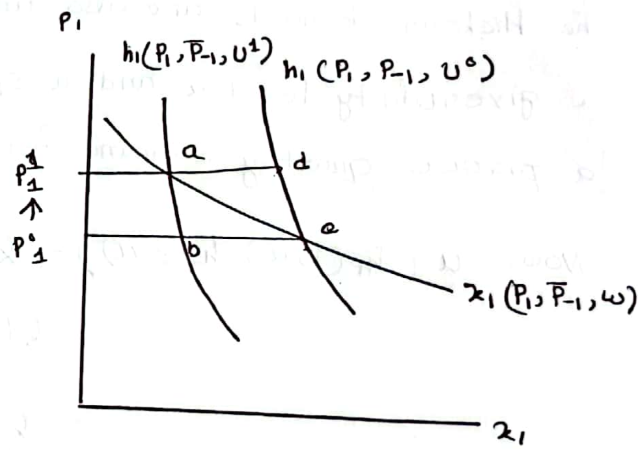
$CS = P_1^0 P_1^1 ea$,,

$CV = P_1^0 P_1^1 ba$,,

so graphically $EV > CS > CV$

Price increase.

same last তা after price increase.



$EV = P_1^1 P_0^1 ba$

$CS = P_1^1 P_0^1 ea$ $\rightarrow$ $EV < CS < CV$

$CV = P_1^1 P_0^1 ed$

Chapter 2 :

13. Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}^N_+$, then examine the properties satisfied by the Walrasian demand function $x(P,w)$. Examine the situations when the sufficient condition in UMP may be violated.

Consumers maximizes his utility level by selecting a bundle x (where x can be a vector) subject to his budget constraint.

$$\max\ u(x)\ ;\ x \geq 0$$
$$\text{s.t.}\ \ p.x \leq w$$

Weierstrass theorem: For optimizing problems defined on reals, if the objective function is continuous and constraints define a close and bounded set, then the solution to such optimizing problem exists.

If, $p \gg 0$ and $w > 0$ (i.e, if $B_{P,w}$ is closed and bounded) and if $u(\cdot)$ is continuous, then there exists at least one solution to the UMP.

If, preferences are strictly convex, then the solution to UMP is unique.

We denote the solution of the UMP as the argmax of the UMP (the optimization problem) and we denote it as argument, x, that solves the walrasian demand correspondence, $x(P,w)$, where $x(P,w)$ is the walrasian demand correspondence, which specifies a demand of every good in $\mathbb{R}^L_+$ for every possible price vector p and every possible wealth level w.

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}^L_+$, then the Walrasian demand $x(P,w)$ satisfy,

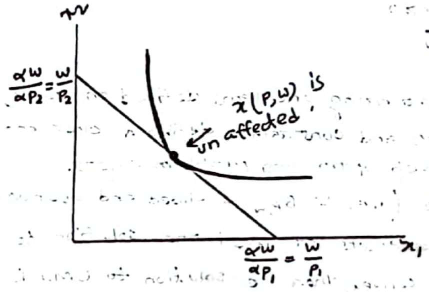
1. Homogeneity of degree of zero :

$$x(P,w) = x(\alpha P, \alpha w) \text{ for all } P, w \text{ and for all } \alpha > 0,$$

That is, the budget set is unchanged.

$$\{x \in \mathbb{R}^L_+ : P.x \leq w\} = \{x \in \mathbb{R}^L_+ : \alpha P.x \leq \alpha w\}$$

Note that we don't need any assumption on the preference relation to show this: We only rely on the budget set being unaffected.



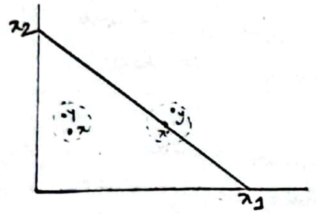
— The preference relation can be linear, and homogeneous at degree of zero (0) and still hold.

2. Walras's Law:

$$P.x = w \text{ for all } x = x(P,w)$$

It follows from LNS, (the property of local nonsatiation (LNS) states that, for any bundle of goods, there of consumer preferences is always another bundle of goods arbitrarily close that is strictly preferred to it) if the consumer selects a Walrasian demand $x \in x(P,w)$, where $p.x < w$, then it means we can still find other bundle y, which is ϵ-close to x, where consumer can improve his utility level.

If the bundle the consumers chooses lies on the budget line, $p \cdot x' = w$, we could then identify bundles that are strictly preferred to x', but these bundles would be unaffordable to the consumer.



For $x \in x(p,w)$, there is a bundle y ϵ-close to x, such that $y \succ x$.
Then, $x \notin x(p,w)$.

3. Convexity / Uniqueness:

a) If the preferences are convex, then the walrasian demand correspondence $x(p,w)$ defines a convex set, a continuum of bundles are utility maximizing.

b) If the preferences are strictly convex, then the walrasian demand correspondence $x(p,w)$ contains a single element.

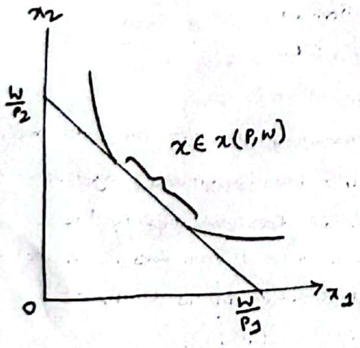


Fig: Convex Preferences

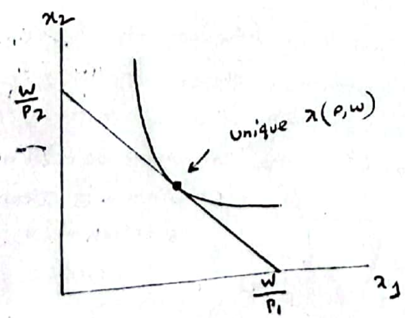


Fig: Strictly convex preferences.

13. Examine the situation, when sufficient condition in UMP may be violated. [For details, see book's page 83-84]

The sufficient condition in UMP may be violated for a max if,

1) $u(x)$ is quasiconcave, convex upper contour set (ucs) and monotone.

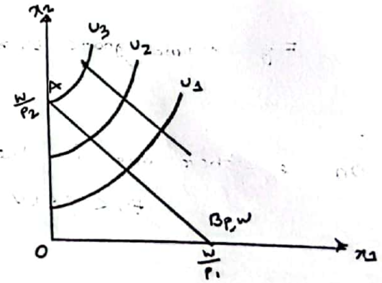
2) The vector of first order derivatives different from zero for all x, that is $\nabla u \neq 0$, for all x.

$\nabla u(x) \neq 0$ for $x \in \mathbb{R}^L_+$

[if $\nabla u(x) = 0$ for some x, then we would be at blissing point (top of the mountain) which violates both LNS and monotonicity]

If,

1) $u(\cdot)$ is non-montone:



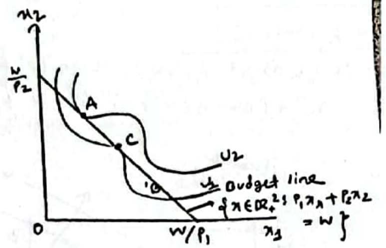
The first condition stating that, the utility function should be monotone only implies that, if we increase both goods simultaneously, we reach a higher utility level, and if we increase the amount of some goods, the consumer is not made worse off, which expeted in most economic applications. If instead, the utility function does not satisfy monotonicity, the consumers indifference curve look like the figure.

— The consumer chooses bundle A (at a corner) since it yields the highest utility level given his budget constraint.

— But, at point A, the tangency condition $MRS_{1,2} = \frac{p_1}{p_2}$ does not hold. Since the consumer would like to further decrease his consumption of x_1.

2) $u(\cdot)$ is not quasiconcave:

Second the condition the first order derivatives are different from zero simply guarantees that there are no bliss points.



- The upper contour sets (UCS) are not convex.
- $MRS_{1,2} = \frac{p_1}{p_2}$ is not a sufficient condition for a max.
- A point of tangency (C) gives a lower utility level than a point of non tangency (B).
- True maximum is at point A.

[For details see book's page 94-99]

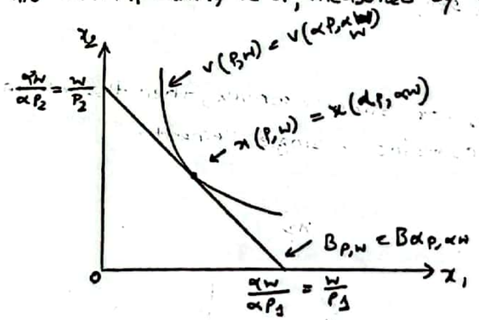
14. Define indirect utility function, Examine it's properties.

The walrasian demand function, $x(p,w)$ is the solution to the UMP. But, the question is what would be the utility function evaluated at the solution of the UMP, ie $x(p,w)$.

This is the indirect utility function, the highest utility level, $v(p,w) \in \mathbb{R}$, associated with UMP. This is the value function of this optimization problem.

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^L$, then the indirect utility function $v(p,w)$ satisfies:

1) Homogenous of degree zero: Increasing p and w by a common factor $\alpha > 0$ does not modify the consumer's optimal consumption bundle, $x(p,w)$ nor his maximal utility level, measured by $v(p,w)$



This property states that increasing market price and wealth by the same proportion does not modify the consumer's budget set. As a consequence, such and increase does not modify the consumer's optimal consumption bundle $x(p,w)$, nor does it affect the maximal utility that the individual reaches in his UMP, as measured by $v(p,w)$, meaning $v(\alpha p, \alpha w) = v(p,w)$.

2. Strictly increasing in w, and Non increasing in p_k for Every good k:-

This property states that, if we increase the wealth of the individual, we enlarge the set of feasible bundles he can afford and, as a consequence, we allow him to reach an indifference curve associated with a higher utility level. Therefore the maximal utility level that he can obtain is strictly increasing in his wealth, where $v(p,w') > v(p,w)$ because $w' > w$.

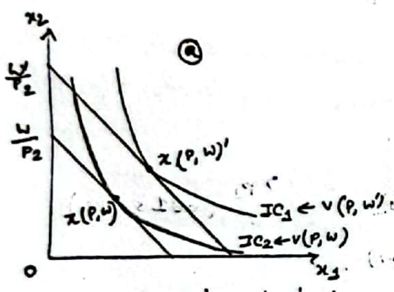


Fig: $v(p,w)$ strictly increasing in w

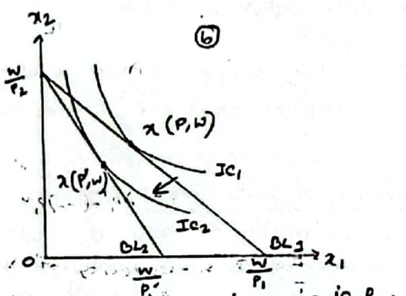


Fig: $V(p,w)$ is nonincreasing in p_k.

In contrast, with an increase in the price of any good, the set of affordable bundles shrinks, so the individual can only reach indifference curves associated with weakly lower utility levels. Hence, an increase in the price of any good k produces a weak reduction in the maximal utility level that the individual can obtain by solving the UMP.

3. Quasi-convexity: The set $\{(p,w) : v(p,w) \le \bar{v}\}$ is convex for any $\bar{v}$.

Interpretation I:

If $(p^1,w^1) \succsim^* (p^2,w^2)$, then $(p^1,w^1) \succsim^* (\lambda p^1 + (1-\lambda)p^2, \lambda w^1 + (1-\lambda)w^2)$; ie, if $A \succsim^* B$, then $A \succsim^* C$.

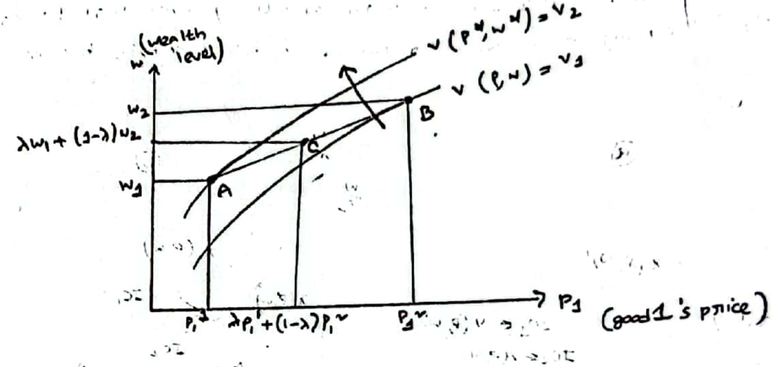


Fig: Quasi-convexity of $v(p,w)$.

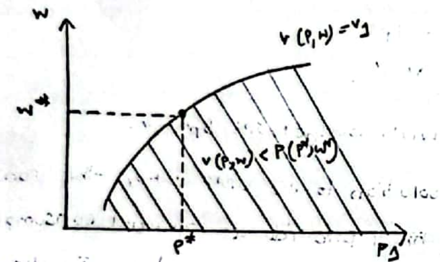
The figure depicts the indirect utility function with the price of good 1 increases from p_1^1 to p_1^2, this individual's wealth must also increase in order to keep his utility level unaffected. In contrast, lower prices and higher wealth levels are associated with higher marginal utilities. Graphically moving to the north-west, entails a higher utility level, $v(p^*,w^*) > v(p,w)$, where $p^* < p$ and $w^* > w$.

Now, for the quasi-convexity of $v(p,w)$, If the maximum utility associated with a given pair of prices and wealth is weakly higher than the maximum utility associated with another pair of prices and wealth, (B), then the convex combination of points A and B that is C, yields a maximum utility that is weakly lower than that associated with A.

That is, if the consumer faces an intermediate price level $

$\lambda p_1^0 + (1-\lambda)p_1''$ and an intermediate weath level $\lambda w_1 + (1-\lambda) w_2$, the maximum utility level that he will be able to reach by solving UMP is lower than, that ~~at point~~ point $A \equiv (p_1^0, w_1)$

Interpretation II : $v(P,w)$ is quasiconvex if the set of (P,w) pairs for which $v(P,w) < v(P^*,w^*)$ is convex.



An alternative interpretation of quasi-convex simply states that the pair of prices and wealth for which the consumer reaches a lower utility level than that under pair (P^*, w^*) defines a convex set. Figure illustrates this property by first depicting the utility level associated with the price - wealth pair (P^*, w^*), $v(P^*, w^*)$ and then shading the set of price weath pairs that yield a lower utility level than $v(P^*, w^*)$ does, thus yielding a convex set.

Interpretation III : Using x_1 and x_2 in the axis, perform following ~~sets~~ steps:

a) When Individuals faces Budget set $B_{p,w}$, his optimal consumption bundle is $x(p,w)$

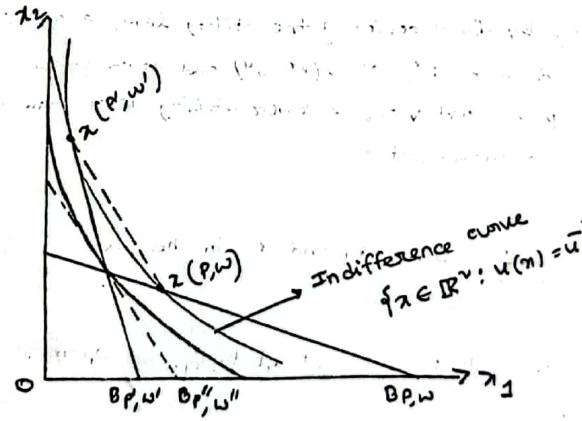
b) When prices and wealth change to p' and w', he faces budget set $B_{p',w'}$, and select bundle $x(p',w')$.

c) Both bundles $x(p,w)$ and $x(p',w')$ induce the same utility $\bar{u}$ because they lie on the same indifference curve, so the indirect utility function satisfies $v(p,w) = v(p',w') = \bar{u}$. We can construct a convex combination of prices and wealth:

$$p'' = \alpha p + (1-\alpha)p'$$
$$w'' = \alpha w + (1-\alpha) w'$$

Which provides us with budget set $B_{p'',w''}$.

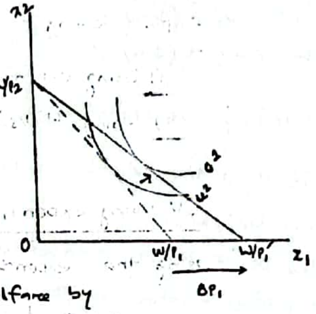
d) Finally, any solution to the UMP when the consumer faces budget set $B_{p'',w''}$, must provide an optimal consumption bundle that lies on a lower indifference curve than $\bar{u}$. Hence the combination of two price - wealth pairs must yield a utility level lower than that from either of the price-wealth pairs alone.



16. a) Distinguish between CV and EV. [For details, see book's page 164-170]
b) How do you measure the welfare effects of a price change.

We can measure the welfare effects that can arise from a price increase or decrease, or alternatively from effects of introducing a tax or subsidy that modifies the price of some good.

Let's consider a price decrease from p_1^0 to p_1^1. We can't compare u^0 to u^1. Instead, we will find a money-metric measure of the consumer's welfare change due to price change.



For change in the price of good 1 from p_1^0 to p_1^1, we can evaluate the change in consumer welfare by using these.

1. **Compensating Variation (CV)**: This welfare responds to the question that how much money would a consumer be willing to give up after a reduction in price to be just as well off as before the price decrease. For this reason it refer to compensating variation, since it represents the change in income that would exactly compensating the consumer for the impact that the price change has on his welfare. This measure maintains the initial utility level before any price change, u^0.

2. **Equivalent Variation (EV)**: It responds to the questions that how much money would a consumer need before a reduction in prices to be as well as as the price decrease. For this reason we refer this measure as the equivalent variation. Since it reflects the change in income that would be equivalent to the impact that the price change has on his welfare. In this case we focus on the final utility level that the individual can reach after the price change, u^1.

There are two approaches to measuring the welfare effect of a price change:

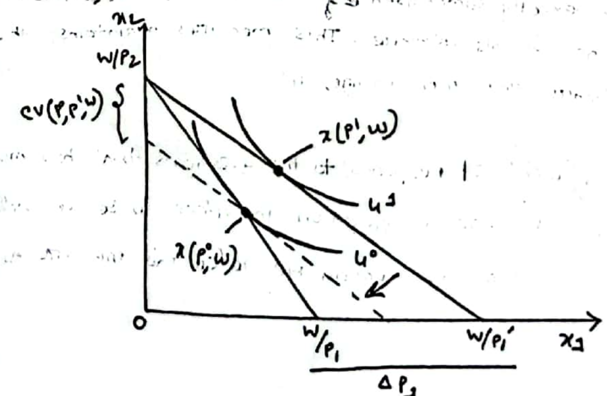
1) Using expenditure function
2) Using Hicksian demand

CV using expenditure function :-

Using the expenditure function we have, $CV(p^0, p^1, w)$ using $e(p, u)$

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$$

This expression represents the amount of money that the consumer is willing to give up after the price decreases (when the price level is p^1 and his utility level has improved to u^1), to be just as well off as before the price decrease (when reaching the lower utility level u^0).



Here's a basic procedure on how to find CV from price decrease. Before the price change, the consumer selects optimal bundle $x(p^0,w)$, reaching utility level u^0 under budget set $B_{p^0,w}$. After p^1 decreases, the consumer chooses an optimal bundle $x(p^1,w)$, under budget set $B_{p^1,w}$. We then adjust the consumer's wealth to make him just as well off as before the price change, by parallel shift in Budget line $B_{p^1,w}$ to the dashed budget line where parallel displacement implies that we maintain the final price ration p^1 but reduce the wealth. Hence the individual can reach his utility (initial) level u^0 at the final price ratio. At this point his expenditure is $e(p^1,u^0)$. Hence the difference in expenditure represents the compensating variation resulting from the price change.

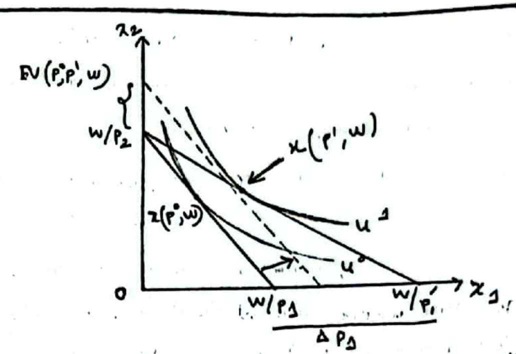
Difference in expenditure: $CV(p^0,p^1,w) = \underbrace{e(p^1,u^1)}_{\text{At } Bp^1,w} - \underbrace{e(p^1,u^0)}_{\text{dashed line}}$.

EV using Expenditure function:

$EV(p^0,p^1,w)$ using $e(p,u)$:-

$EV(p^0,p^1,w) = e(p^0,u^1) - e(p^0,u^0)$

The EV represents the amount of money that the consumer is willing to receive instead of the price decrease. That is, at the initial price level p^0 (when his utility level is still u^0) we find the monetary amount that makes him as well off as after the price decrease, (when he reaches the higher utility level u^1).



There is a basic procedure on how to find the EV from a price decrease. Before price change, the consumers selects the optimal bundle $x(p^0,w)$ reaching utility level u^0. After p_1 decreases, the consumer chooses an optimal bundle $x(p^1,w)$ under budget set $B_{p^1,w}$. Using the definition of the EV, we now must increase the initial wealth of the individual (before the price change), $B_{p^0,w}$, to make sure he is as well of as after the price change. Graphically we do so by upward shifting the initial budget line in a parallel fashion. Intuitively, we are increasing the initial wealth of the individual so that, at the initial price ratio, he can still reach utility level u^1 (the high utility level he can obtain after the price decrease. So, the difference in expenditure:

$EV(p^0,p^1,w) = \underbrace{e(p^0,u^1)}_{\text{dashed line}} - \underbrace{e(p^0,u^0)}_{\text{At } B_{p^0,w}}$

That is, EV represents the amount of money needed to obtain the new utility at the old price.

~~CV using~~

CV using Hicksian Demand :

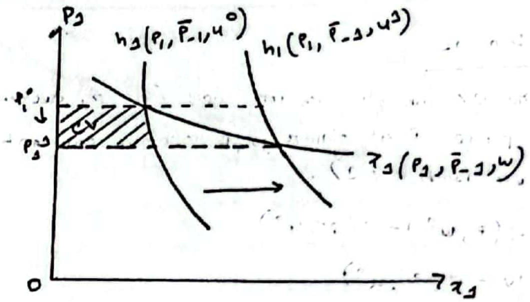
The expression of the CV can also rearranged (as p_1 decreases from p_1^1 to p_1^0, while the price of all other goods remains constant, in that $p_k^0 = p_k^1$ for all other goods $k \neq 1$) ;

$$CV(p^0, p^1, w) = e(p', u^1) - e(p^1, u^0)$$
$$= w - e(p^1, u^0)$$
$$= e(p^0, u^0) - e(p^1, u^0)$$
$$= \int_{p_1^0}^{p_1^1} \frac{\partial e(p_1, \bar{p}_{-1}, u^0)}{\partial p_1} dp_1$$
$$= \int_{p_1^0}^{p_1^1} h_1(p_1, \bar{p}_{-1}, u^0) dp_1$$

Since $e(p^1, u^1) = e(p^0, u^0) = w$ and $\partial e(p,u)/\partial p_1 = h_1(p_1, u)$, and let $\bar{p}_{-1}$ denote the price vector of all other goods $k \neq 1$ such that $\bar{p}_{-1} = (p_2, p_3, \dots p_L)$, which are unaffected by the price change. Since $p_1^0 > p_1^0$, the CV of a price decrease is negative, thus indicating the amount of money that the individual would willing to give up after the price decrease in order to be as well off as before the price decrease.

Here the case is,

- Normal good
- Price decrease.



Graphically, CV is represented by the area to the left of the Hicksian demand curve for good 1 associated with the utility level u^0, and lying between price p_1^1 and p_1^0. The welfare gain is represented by the shaded region.

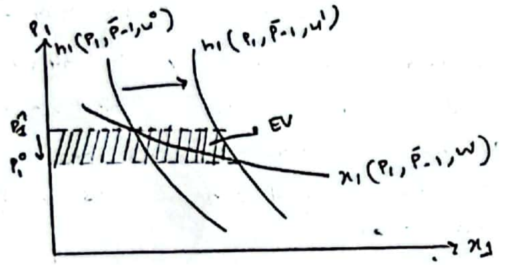
EV using Hicksian Demand :

If p_1 decreases, meaning $p_1^1 < p_1^0$ and $p_k^1 \neq p_k^0$, for all other goods $k \neq 1$,

The EV is,

$$EV(p^0, p^1, w) = e(p^0, u^1) - \underbrace{e(p^0, u^0)}_{w}$$
$$= e(p^0, u^1) - w$$
$$= e(p^0, u^1) - \underbrace{e(p^1, u^1)}_{w}$$
$$= \int_{p_1^1}^{p_1^0} \frac{\partial e(p_1, \bar{p}_{-1}, u^1)}{\partial p_1} dp_1 = \int_{p_1^0}^{p_1^1} h_1(p_1, \bar{p}_{-1}, u^1) dp_1,$$

Here as a normal good and price decrease. EV represented by the area to the left of the Hicksian demand curve, associated with utility level u^1, and lying between the initial price p_1^0 and the final price p_1^1



The definition of the EV emphasizes that we need to adjust the initial wealth of the individual such that, under the initial price ratio, he can reach his final utility level u^1 and remain at $h_1(p_1, \bar{p}_{-1}, u^1)$. His welfare gain is measured by the shaded area in the figure.

17. Graphically show that for a decrease in the price of a commodity.
(when both goods are normal)
EV > CS > CV

[For details see book's page 174 - 177]

and for price increase,
EV < CS < CV

The walrasian demand is easier to observe than the Hicksian demand. If we consider a walrasian demand, a decrease in p would imply an increase in consumer surplus (CS)

Price decrease for Normal good :

Let's consider a cobb douglas utility function $u(x,y) = (x^{1/2})(y^{1/2})$, a wealth level of $w = \$100$. The price of good y normalized to \$1. If the price of good x is p_x, it's walrasian demand is given by

given by $x(p_x, 100) = 50/p_x$.

Now, assume that the price of good x decreases from \$4 to \$3. The increase in consumer surplus is given by CS = 34.64.

In order to measure the CV, we first find the initial bundle $x(p_0, w)$, the final bundle $x(p_1, w)$, and bundle $x(p_1, w')$ where w' denotes the adjusted income so the individual can reach his initial utility level at the final price ratio.

It show that,
$x(p_0, w) = (25, 50)$
$x(p_1, w) = (50, 50)$
$x(p_1, w') = (35.35, 35.35)$

Hence CV is given by the difference in income that the individuals needs to purchase these two bundles, as follows,

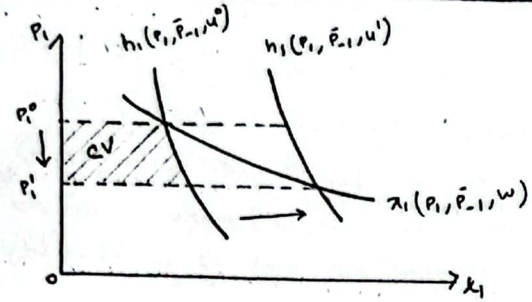
CV = 100 - 70.71 = 29.29.

Last to measure the EV, we need to obtain bundle $x(p_0, w'')$ where w'' represents the adjusted weath so the individual can reach his final utility level before the price change. Here, ~~x(p0, w'') = 3~~ $x(p_0, w'') = (35.35, 70.71)$. As a consenquence, the EV is given the difference in incomes from purchasing these two bundles.

EV = 141.42 - 100 = 41.42.

In summary, EV > CS > CV

(a)



(b)

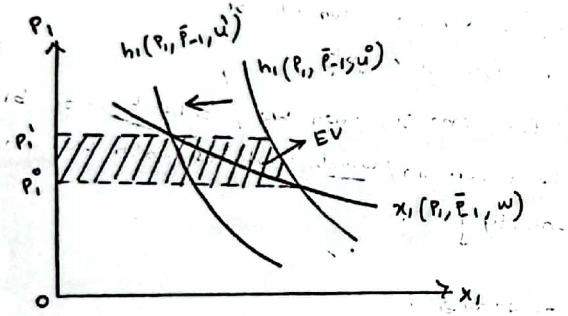


Fig: Normal goods, $EV > CS > CV$ for price decreases.

On the other hand, for the price increases for normal good.

$EV < CS < CV$